

Fatma Adel Mohamed

Entry-Level Backend Developer (Java & Spring)

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 leetcode  codeforces(Fatma_3del.)  codeforces(YILDIZ)

Professional Summary

Computer Science graduate with hands-on experience building scalable backend systems using Java and Spring Boot. Skilled in Microservices, REST APIs, Spring Security, JWT Authentication, Redis, Docker, AI integration, RAG, LangChain4j, OpenAI APIs, and database design. Passionate about backend development, clean architecture, and building secure, high-performance applications.

Experience

Backend Bootcamp Trainee, Spring Boot Bootcamp  2025

- Completed intensive backend development training using Java, Spring Boot, Spring MVC, Hibernate and Angular.
- Built multiple backend applications applying layered architecture, REST APIs, Spring Security and Spring Data JPA.
- Worked with MySQL, Maven, Git, Swagger, Docker and Postman.
- Applied software engineering best practices, object-oriented programming and clean architecture.

Education

Bachelor's Degree in Computer Science, Al-Obour High Institute 2022 – 2026
GPA: Excellent Cairo
Graduation Project (ZooVerse -- Integrated Zoo Management System): Excellent

Technical Skills

Languages Java, Python, C++, SQL, JavaScript	Backend Spring Boot, Spring MVC, Spring Security, Java EE, Microservices
AI / LLM OpenAI API, LangChain4j, RAG, ChromaDB, Vision AI, Prompt Engineering	Databases MySQL, Redis, JDBC, Hibernate ORM, Spring Data JPA
Security / APIs REST APIs, JWT, Refresh Tokens, OAuth2, Email OTP, Swagger	Tools Docker, Git, GitHub, Maven, IntelliJ IDEA, VS Code, Postman, WebSocket
Concepts OOP, Data Structures, Algorithms, MVC, SOLID Principles, Clean Architecture	

Projects

HealthTrack Pro -- AI Personal Health Management System [🔗](#), *Spring Boot, Microservices*

- Collaborated in building an AI-powered personal health management platform using Spring Boot Microservices.
- Developed health tracking modules for Blood Pressure, Glucose, Pulse, Weight and Body Temperature monitoring.
- Integrated OpenAI, LangChain4j, RAG and ChromaDB to deliver intelligent healthcare recommendations.
- Implemented Vision AI for medical image and report analysis.
- Secured the platform using JWT Authentication, Refresh Tokens, Google OAuth2 and Email OTP verification.
- Improved performance using Redis caching, Docker containerization and WebSocket real-time notifications.

Animal AI Chatbot [🔗](#), *Spring Boot, OpenAI*

- Developed an AI chatbot using RAG architecture and vector search.
- Integrated Vision AI and multilingual support.
- Applied modular architecture and clean code principles.

ToDo App Backend [🔗](#), *Spring Boot, Microservices*

- Engineered a microservices-based task management application consisting of User Service and Todo Service.
- Secured the system using JWT Authentication and Email OTP verification.
- Developed complete CRUD operations.
- Connected services through REST APIs with independent MySQL databases.
- Documented APIs using Swagger and followed layered architecture with Spring Data JPA.

CRUD Management System [🔗](#), *Spring MVC*

- Developed a CRUD web application using Spring MVC, Hibernate ORM and MySQL.
- Applied MVC architecture, layered design and Hibernate for data persistence.

Achievements

Competitive Programming

Solved 1,000+ algorithmic problems across Codeforces and LeetCode, strengthening problem-solving, data structures and algorithmic thinking.

Languages

Arabic

Native

English

Professional Working Proficiency